Sahaj Bhadja

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Summary

Backend Engineer with 2+ years building **Python/Java microservices**, **distributed systems**, and secure **cloud-native data platforms**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms by optimizing indexed queries and data models for a **multi-tenant Azure SQL** platform supporting 500+ users with row-level isolation controls.
- Achieved 99%+ reliability across 5K+ monthly jobs by building **fault-tolerant FastAPI pipelines** with incremental processing, validation, reconciliation, error isolation, cloud storage, and stable repeatable CI/CD-backed production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python service** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD at scale.
- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive workflows under concurrent production load and model execution.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Increased data-pipeline efficiency by 68% using **AWS Glue and S3**, automating builds and 20+ unit/integration suites through Jenkins, PyTest, Mockito, GitHub, nightly reporting, and automated failure diagnostics.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **microservice-based CRM platform** covering authentication, permissions, customers, billing, resources, leads, and notifications with Node.js, WebSockets, and Azure SQL across distributed locations.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT controls, Azure monitoring, automated tests, and Docker/Jenkins deployments across modular services, distributed locations, and production releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered sub-200ms **Flask APIs** across 50K+ booking records, using Firebase transactions, real-time synchronization, and idempotent booking workflows for mobile, web, pricing, and time-slot experiences across client platforms.
- Reduced manual refund handling by 70% through **payment webhook automation**, integrating disputes, JWT access, and hourly Databricks/Spark pipelines while aligning end-to-end architecture with customer requirements and operational reporting.

Skills

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, Bash
Backend & Architecture: Microservices, Distributed Systems, FastAPI, Flask, Node.js, REST APIs, Data Modeling, Multi-Tenancy, Fault Tolerance
Cloud & Data: AWS, Azure, GCP, PostgreSQL, Azure SQL, AWS RDS, AWS S3, AWS Glue, MongoDB
Security & Reliability: Data Protection, RBAC, JWT, Row-Level Security, Data Validation, Audit Logging, Observability
Platform Engineering: Kubernetes, Docker, CI/CD, GitHub Actions, Jenkins, Git, PyTest, Performance Tuning

Projects

Privacy Signal Orchestrator - Design Concept

- Proposed a **Python microservice architecture** for ingesting consent events, resolving identities, enforcing purpose-based permissions, and propagating idempotent decisions through PostgreSQL-backed workflows with retries, audit logs, and observability.

Distributed Data Rights Pipeline - Design Concept

- Proposed **Kubernetes-based workers** coordinating access and deletion requests across heterogeneous datastores, with tenant isolation, checkpointed execution, failure recovery, encrypted payloads, reconciliation, and end-to-end completion receipts with auditable evidence.

FinNexus | Live

- Built a **Kubernetes-deployed data platform** supporting 50K+ users and 1M+ transactions with secure Plaid APIs, MongoDB, WebSockets, token-based controls, real-time synchronization, filtering, and automated delivery workflows reliably.